

Calibrating mixed-model association under population structure: exact LOCO, two-step statistics and a structure-aware denominator

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Abstract

Mixed-model association scans now test each SNP against a kinship built without its own chromosome (leave one chromosome out, LOCO), and the methods used for large cohorts compute such scans in two steps. They scale a retrospective score by a single genome-wide calibration constant. We describe mixmogam 2, which implements exact LOCO EMMAX, BOLT-LMM-inf, BOLT-LMM and LDAK-KVIK on a shared streaming LOCO operator, and we evaluate calibration per SNP rather than genome-wide. On 1,307 *Arabidopsis thaliana* RegMap accessions with a null chromosome, single-constant statistics are anti-conservative for SNPs that load weakly on the leading kinship eigenvectors and conservative for SNPs that load strongly on them. The reference LDAK-KVIK program gives $\lambda_{GC} = 1.21$ and a false-positive rate of 1.67% at $\alpha = 1\%$ in the lowest loading quintile, against 0.92 and 0.78% in the highest, while its genome-wide λ_{GC} is 1.04. A denominator built on each SNP's own LOCO eigenbasis restores flat calibration (1.04 and 1.03). In simulation, a kinship alone absorbs a strong environment that tracks ancestry only partly: exact LOCO gives $\lambda_{GC} = 1.27$, and null SNPs in the top 1% of loading on the environment reach mean $\chi^2 = 2.27$. One principal component as a fixed covariate removes the residual (1.11 and 1.13). The non-LOCO scan's apparent calibration (0.98) is the same residual offset by proximal-contamination deflation. We also describe two traps in simulation design, a confounder built from the tested SNPs and LD blocks cut through LD, and their remedies.

1 Introduction

Linear mixed models are the standard tool against confounding by population structure and relatedness in genome-wide association studies (Kang et al., 2008, 2010; Yang et al., 2010; Lippert et al., 2011; Zhou and Stephens, 2012; Yang et al., 2014). The confounders are the environment and the genetic background, both of which track ancestry (Vilhjálmsen and Nordborg, 2013); principal components (Price et al., 2006) and mixed models (Price et al., 2010; Sul et al., 2018) address them in different ways. Two refinements are now routine. First, testing a SNP against a kinship that contains it lets the polygenic term absorb part of the SNP's own effect (proximal contamination), so scans leave the tested chromosome out of the kinship (Listgarten et al., 2012; Yang et al., 2014). Second, for large cohorts the exact statistic is replaced by a two-step one. The first step fits a LOCO polygenic model without forming the kinship; the second tests each SNP against the LOCO residual with a retrospective score, scaled by one genome-wide constant (Loh et al., 2015; Mbatchou et al., 2021; Hof and Speed, 2025).

One constant is exact only if every SNP's prospective denominator is proportional to its squared norm (Section 2.2). Under population structure it is not, because a SNP aligned with the leading kinship eigenvectors has a smaller prospective denominator than an unaligned SNP of equal norm. The BOLT-LMM authors tested their method on human data only, and noted that treating this

denominator as nearly constant may fail in plant and animal data (Loh et al., 2015). A genome-wide λ_{GC} averages over SNPs and need not show the problem. Our estimand is therefore the null distribution of each SNP’s statistic given its loading on the leading kinship eigenvectors, summarized by λ_{GC} and the false-positive rate (FPR) within loading quintiles.

This report makes four contributions:

1. an implementation of exact LOCO EMMAX and of the two-step statistics of BOLT-LMM and LDAK-KVIK on one streaming LOCO operator (Section 2);
2. per-SNP calibration on real *A. thaliana* genotypes (Horton et al., 2012) and on simulated structure, including the reference LDAK-KVIK program (Sections 3.1 and 3.2);
3. a structure-aware denominator that restores per-SNP calibration for BOLT-LMM-inf (Section 2.3);
4. a simulation study with a confounder that is independent of the tested SNPs and with LD-split blocks. It compares LOCO with non-LOCO, with and without a principal-component covariate (Sections 3.3 and 3.4).

mixmap’s first version provided the multi-locus mixed model (Segura et al., 2012); the present engine replaces it. Full derivations and pseudocode are in the technical documentation, `docs/methods.pdf`.

2 Methods

2.1 Model and exact LOCO scan

Let n be the number of samples, $\mathbf{X} \in \mathbb{R}^{n \times q}$ the covariates (with an intercept) and $\mathbf{K}$ a genomic relationship matrix built from standardized genotypes (Yang et al., 2010). The model is

$$\mathbf{y} = \mathbf{X}\beta + u + e, \quad u \sim \mathcal{N}(0, \sigma_g^2 \mathbf{K}), \quad e \sim \mathcal{N}(0, \sigma_e^2 \mathbf{I}_n), \quad (1)$$

with $\delta = \sigma_e^2 / \sigma_g^2$ and $\mathbf{V} = \sigma_g^2 (\mathbf{K} + \delta \mathbf{I}_n)$. Variance components are fitted by REML from one eigendecomposition of $\mathbf{K}$ (Kang et al., 2008). Write $\mathbf{H} = \mathbf{V}^{-1/2}$, $\mathbf{Q}$ for an orthonormal basis of $\mathbf{H}\mathbf{X}$, and $r = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{H}\mathbf{y}$. For a SNP g with $\tilde{g} = (\mathbf{I} - \mathbf{Q}\mathbf{Q}^\top)\mathbf{H}g$, the EMMAX statistic (Kang et al., 2010) is

$$F(g) = \frac{(\tilde{g}^\top r)^2 / \tilde{g}^\top \tilde{g}}{(r^\top r - (\tilde{g}^\top r)^2 / \tilde{g}^\top \tilde{g}) / (n - q - 1)} \sim F_{1, n-q-1}. \quad (2)$$

All SNPs of a block are transformed by matrix products, so the scan is dense linear algebra without a per-SNP solve. For LOCO, SNPs are grouped by chromosome (more than 25 chromosomes are merged into 25 contiguous groups), and the kinship of group g is obtained by subtraction,

$$\mathbf{K}_{-g} \propto \sum_j \mathbf{z}_j \mathbf{z}_j^\top - \sum_{j \in g} \mathbf{z}_j \mathbf{z}_j^\top, \quad (3)$$

with $\mathbf{z}_j$ the globally standardized genotypes and the result rescaled to mean diagonal one. The variance components are refitted for each group, and the SNPs of the group are scanned with (2).

2.2 Two-step statistics and the constant-ratio assumption

BOLT-LMM-inf (Loh et al., 2015) estimates (σ_g^2, δ) once and solves $\mathbf{w}_g = (\mathbf{K}_{-g} + \delta \mathbf{I})^{-1} \mathbf{y}$ for every group by conjugate gradients, never forming $\mathbf{K}_{-g}$. A SNP $\mathbf{z}_j$ in group g gets the retrospective statistic

$$\chi_j^2 = \frac{(\mathbf{z}_j^\top \mathbf{w}_g)^2}{c (\mathbf{z}_j^\top \mathbf{z}_j / n) \|\mathbf{w}_g\|^2}, \quad (4)$$

where the constant c is the mean, over 30 random SNPs, of the ratio that makes (4) equal the prospective statistic $(\mathbf{z}_j^\top \mathbf{V}_{-g}^{-1} \mathbf{y})^2 / \mathbf{z}_j^\top \mathbf{V}_{-g}^{-1} \mathbf{z}_j$. One constant can match the prospective statistic for every SNP only if

$$\mathbf{z}_j^\top (\mathbf{K}_{-g} + \delta \mathbf{I})^{-1} \mathbf{z}_j \propto \mathbf{z}_j^\top \mathbf{z}_j \quad \text{for all } j. \quad (5)$$

Write $\mathbf{K}_{-g} = \mathbf{U} \Lambda \mathbf{U}^\top$. The left side of (5) is $\sum_k (\mathbf{U}_k^\top \mathbf{z}_j)^2 / (\lambda_k + \delta)$, so a SNP that loads on the large leading eigenvalues (structure) has a small denominator, and a SNP that loads on the bulk has a large one. A constant fitted to the average is then too small for unaligned SNPs (inflation) and too large for aligned ones (deflation). BOLT-LMM (Loh et al., 2015) replaces $\mathbf{w}_g$ by the residual of a two-Gaussian mixture prior fitted by variational Bayes and matches the LD Score regression intercept (Bulik-Sullivan et al., 2015). LDAK-KVIK (Hof and Speed, 2025) uses an elastic-net prior and a scaling rule that is triggered by a structure test. Both keep one constant. mixmogam’s re-implementations and their departures from the reference programs are documented in `docs/methods.pdf`.

2.3 Structure-aware denominator

mixmogam replaces the constant by a per-SNP denominator on the top- k eigenpairs $(\mathbf{U}_g, \Lambda_g)$ of the SNP’s own LOCO kinship, with the rest of the spectrum treated as flat at its mean $\bar{\lambda}_g$:

$$D_j = c' \sigma_g^2 \left[\|(\Lambda_g + \delta)^{-1/2} \mathbf{U}_g^\top \mathbf{z}_j\|^2 + \frac{\mathbf{z}_j^\top \mathbf{z}_j - \|\mathbf{U}_g^\top \mathbf{z}_j\|^2}{\bar{\lambda}_g + \delta} \right], \quad \chi_j^2 = \frac{(\mathbf{z}_j^\top \mathbf{w}_g)^2}{D_j}, \quad (6)$$

with c' calibrated on the same 30 SNPs. The basis must come from the LOCO kinship: a basis of the full kinship contains each SNP’s own LD block and reintroduces proximal contamination into the denominator. The LOCO eigenbases of all groups are computed together by randomized subspace iteration through one streaming operator. The archived calibration runs used $k = 64$; the current default doubles k from 64 to at most 512 until the spread of c'_j over the calibration SNPs is below 3%. For the mixture and elastic-net statistics, the per-SNP factor of the infinitesimal model is applied before their own calibration; this transfer is a heuristic.

2.4 Principal components under strong environmental stratification

Under (1) the variance of $\mathbf{y}$ along an eigenvector of $\mathbf{K}$ is $\sigma_g^2 \lambda_k + \sigma_e^2$, the same function of the eigenvalue for every direction. An environment that moves the phenotype along the leading (ancestry) eigenvector by a large amount therefore cannot be represented, and REML compensates by inflating σ_g^2 . The standard remedy adds the top principal components of $\mathbf{K}$ to $\mathbf{X}$ (Price et al., 2006, 2010); we evaluate it with one component.

2.5 Implementation

The two-step methods hold standardized, covariate-projected SNP blocks that never straddle a LOCO group. One pass over the blocks applies every $\mathbf{K}_{-g}$ to its own column of a matrix, so the

LOCO solves, the calibration solves and the LOCO eigenbases share matrix products. Variational Bayes for the mixture priors runs all cross-validation folds or all LOCO groups at once. Variance components for large n come from deflated stochastic Lanczos quadrature on the streaming operator. Scans default to single precision and fits to double precision. A fitted model and its cached fit must not reference each other: such a cycle kept every LOCO group’s kinship and eigendecomposition in memory until Python’s cyclic garbage collector ran, about 8 GB per exact LOCO scan at $n = 4,000$. The model now stores the fit without the back reference.

2.6 Evaluation designs

Per-SNP calibration. The genetic value is built from every chromosome except the last: a polygenic component with $h^2 = 0.4$ on standardized genotypes plus 10 QTLs with $h^2 = 0.1$. The SNPs on the last chromosome are then null, and their LOCO kinship is the generating one. Each null SNP is assigned the share of its squared norm that falls in the top 10 eigenvectors of the kinship, and null SNPs are binned into quintiles of this loading. We report λ_{GC} (median χ^2_1 over 0.455) and the FPR at $p < 0.01$ per quintile, as means over six replicate phenotypes. The data are (i) the *A. thaliana* RegMap panel (Horton et al., 2012): 1,307 inbred accessions, every fourth SNP with minor allele count at least 10 (53,167 SNPs on 5 chromosomes, 13,209 of them on the null chromosome 5); and (ii) simulated LD-free genotypes, $n = 1,300$ and $m = 20,000$ on 5 chromosomes, without structure and with 4 populations at $F_{ST} = 0.3$. On RegMap the reference LDAK-KVIK program (LDAK version 6, `--kvik-step1` and `--kvik-step2` with default settings) was run on the same phenotypes.

Simulation study. Genotypes come from the coalescent with recombination (Kelleher et al., 2016; Baumdicker et al., 2022): $N_e = 10^4$, recombination and mutation rates 10^{-8} per base pair per generation, minor allele frequency above 1%. Each LD block is its own “chromosome”, so a locus is a block. In scenarios S1 and S3 one simulated segment is cut into $m/200$ blocks of 100–400 SNPs by the optimal LD split of Privé (2022), which minimizes the r^2 leaking across block boundaries (cf. Berisa and Pickrell, 2016). Traits have heritability h^2 , half from an infinitesimal background over all SNPs and half from QTLs: S1 has $n = 800$, $m = 20,000$, $h^2 = 0.5$ and 20 QTLs; S3 has $n = 4,000$, $m = 50,000$, $h^2 = 0.5$ and 30 QTLs. Power and false loci are counted per block at the 5% Bonferroni threshold; false loci are significant blocks without a QTL. Three seeds per scenario.

Confounding by environment (S2). The sample comes from two demes that split 400 generations ago (each and their ancestor of size 10^4), 400 diploids from each. Every one of 100 blocks of 200 SNPs is an independent coalescent replicate, so blocks are unlinked. The trait is the S1 construction with $h^2 = 0.25$ and 10 QTLs, plus an environment equal to the standardized deme indicator, which carries 50% of the liability variance. The environment correlates with every SNP that drifted between the demes, but no tested SNP causes it. Null SNPs are those in blocks without a QTL. We report their λ_{GC} and their mean χ^2 in the bottom half and in the top 1% of loading on the environment (squared correlation with it). As a negative control, the superseded design put the confounder on the leading eigenvector of the tested SNPs’ own kinship in a panmictic sample. That axis is a weighted sum of the tested SNPs, so the SNPs that load on it most carry the confounder themselves.

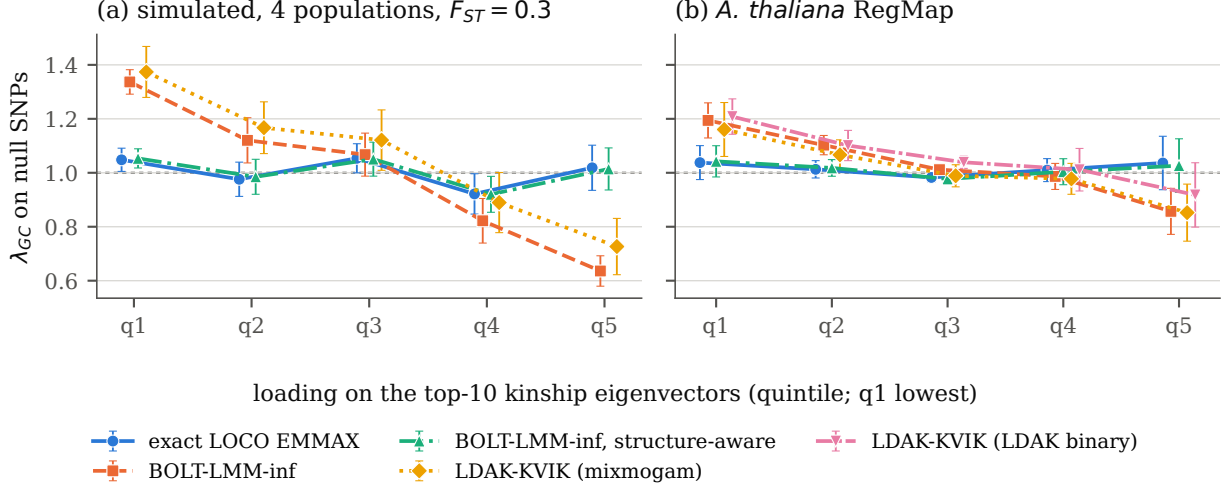


Figure 1: Calibration on null SNPs by loading on the leading kinship eigenvectors. Points are means over six replicate phenotypes, bars 95% intervals over replicates; series are offset horizontally for readability. (a) LD-free genotypes from four populations at $F_{ST} = 0.3$. (b) *A. thaliana* RegMap with a null chromosome, including the reference LDAK-KVIK program run on the same phenotypes.

3 Results

3.1 Per-SNP calibration on *A. thaliana*

Exact LOCO EMMAX is flat across the loading quintiles on RegMap (λ_{GC} 1.04 in the lowest and highest quintile, FPR 0.96% and 0.98%; Table 1, Figure 1b). Every single-constant statistic shows a gradient. BOLT-LMM-inf goes from $\lambda_{GC} = 1.19$ (FPR 1.66%) to 0.86 (0.56%), mixmogam’s LDAK-KVIK from 1.16 (1.44%) to 0.85 (0.49%), and the reference LDAK-KVIK program from 1.21 (1.67%) to 0.92 (0.78%). Their genome-wide λ_{GC} are 0.99–1.04. The reference program detected strong structure in every replicate and scaled its statistics by 1.14–1.39; its per-SNP χ^2 correlates 0.958 with exact LOCO EMMAX and 0.966 with mixmogam’s re-implementation. The structure-aware denominator gives 1.04 and 1.03 in the extreme quintiles (FPR 0.97% and 0.99%) and correlates 0.999 with the exact statistic.

3.2 Simulated structure

Without structure every method is flat (λ_{GC} 0.98–1.04 in every quintile). With four populations at $F_{ST} = 0.3$ the gradient is steeper than on RegMap (Figure 1a): BOLT-LMM-inf goes from 1.34 to 0.64 (FPR 2.45% to 0.11%), BOLT-LMM from 1.35 to 0.70, and LDAK-KVIK from 1.37 to 0.73 (2.92% to 0.11%), while exact LOCO stays between 0.92 and 1.05. The structure-aware denominator brings BOLT-LMM-inf to 1.05 and 1.01 in the extreme quintiles. Each two-step result reports the spread of its calibration ratio over the 30 calibration SNPs, which flags the problem: 0.5% without structure, 23% at $F_{ST} = 0.3$ and 13% on RegMap, against 0.4%, 0.3% and 3% with the structure-aware denominator. Mean χ^2 at the QTLs differed by a few percent across methods.

3.3 Confounding by an environment that tracks ancestry

The two-deme samples have $F_{ST} = 0.020$ (Hudson’s estimator; Bhatia et al., 2013). The leading GRM eigenvalue is 16–17, against about 3 for LD, and PC1 has $r^2 = 0.97$ with deme. The

Table 1: Genome-wide λ_{GC} , λ_{GC} in the lowest (q1) and highest (q5) loading quintile, and the false-positive rate (%) at $p < 0.01$ in those quintiles, for null SNPs. Means over six replicates; nominal FPR is 1%.

method	λ_{GC} , all	λ_{GC}		FPR (%)	
		q1	q5	q1	q5
<i>simulated, 4 populations, $F_{ST} = 0.3$</i>					
exact LOCO EMMAX	1.00	1.05	1.02	1.12	1.11
BOLT-LMM-inf	0.97	1.34	0.64	2.45	0.11
BOLT-LMM-inf, structure-aware	1.00	1.05	1.01	1.14	1.06
BOLT-LMM	1.00	1.35	0.70	2.58	0.11
LDAK-KVIK (mixmogam)	1.02	1.37	0.73	2.92	0.11
<i>A. thaliana RegMap</i>					
exact LOCO EMMAX	1.01	1.04	1.04	0.96	0.98
BOLT-LMM-inf	1.01	1.19	0.86	1.66	0.56
BOLT-LMM-inf, structure-aware	1.01	1.04	1.03	0.97	0.99
LDAK-KVIK (mixmogam)	0.99	1.16	0.85	1.44	0.49
LDAK-KVIK (LDAK binary)	1.04	1.21	0.92	1.67	0.78

environment inflates the plain linear model to $\lambda_{GC} = 11.8$ and makes every null block significant (Table 2). The mixed models remove most of this, but not all.

1. *The confounder no longer favours non-LOCO.* Under the superseded design, null SNPs in the top 1% of loading had mean χ^2 9.8 with LOCO and 6.0 without, which the overall λ_{GC} (1.39 and 0.98) hid (Figure 2a). With an independent confounder, both show the same mild gradient, twice their bottom-half mean (Figure 2b).
2. *A kinship alone absorbs the environment only partly.* REML inflates the pseudo-heritability to 0.66 against a simulated genetic share of 0.125. Null SNPs keep a loading gradient, and exact LOCO sits at $\lambda_{GC} = 1.27$ (BOLT-LMM-inf 1.28). With PC1 as a covariate the gradient vanishes (1.10 in the bottom half, 1.13 in the top 1%) and λ_{GC} drops to 1.11.
3. *Non-LOCO’s 0.98 is not better calibration.* It is the same residual, offset by the proximal-contamination deflation that the non-LOCO scan also shows without structure (S1: 0.95 against 1.25 for LOCO; Table 3).
4. *At the locus level the residual stays below the Bonferroni threshold.* Every mixed model has at most one false locus per scan. Power at $n = 800$ is low for all of them: 6 of 30 QTLs were found with LOCO and 4 without.

3.4 LOCO, power and LD-split blocks

LOCO raises locus-level power in S3 from 0.33 to 0.51 (Table 3). Its λ_{GC} (1.65) exceeds the non-LOCO value (0.91) because the infinitesimal background puts true signal on every SNP. LOCO keeps that signal, while the non-LOCO scan is deflated by proximal contamination. The same background makes some “false” loci real associations; at $n = 4,000$ LOCO detects them.

Cutting the coalescent segment every 200 SNPs, as the superseded design did, splits strong LD. The largest r^2 between the last and first 20 SNPs of neighbouring blocks had median 1.0 and exceeded 0.5 at 99–100% of boundaries, and every S1 false locus under those cuts was a QTL tag

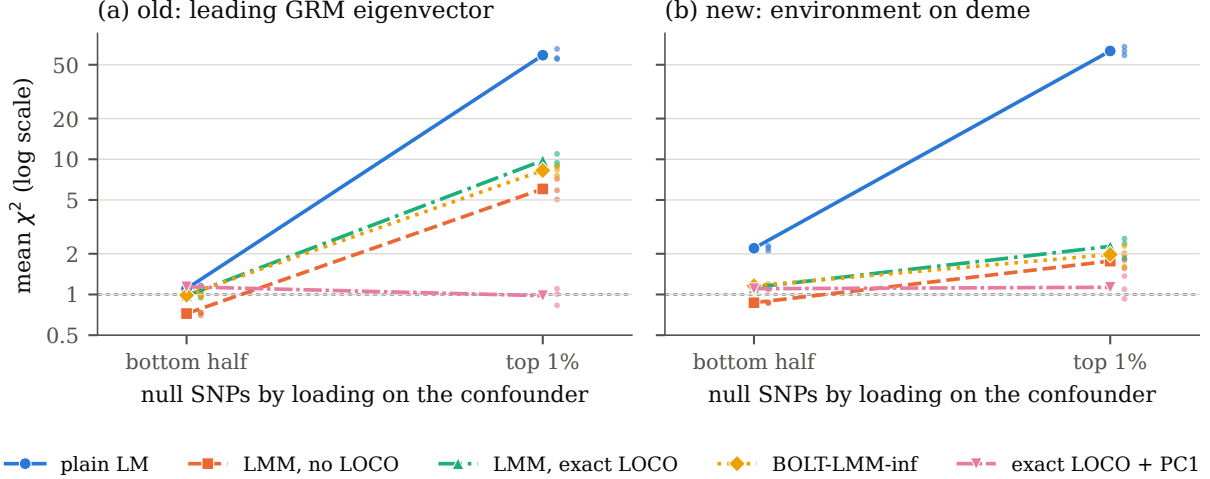


Figure 2: Mean χ^2 of null SNPs in the bottom half and the top 1% of loading on the confounder, for the superseded confounder on the leading eigenvector of the tested SNPs’ kinship (a) and for an environment on deme membership (b). Lines join means over three seeds; small points are the seeds. The dashed line marks $\chi^2 = 1$.

Table 2: Scenario S2 with an environment on deme membership, means over three seeds: λ_{GC} over all SNPs and over null SNPs, mean χ^2 of null SNPs by loading on the confounder (bottom half, top 1%), the same for the superseded design, false loci per scan and locus-level power.

method	λ_{GC}	λ_{GC} , null	null χ^2		superseded		false loci	power
			low	top 1%	low	top 1%		
plain LM	11.84	11.90	2.20	63.24	1.10	58.9	90.33	1.00
LMM, no LOCO	0.98	0.96	0.87	1.77	0.72	6.0	0.00	0.13
LMM, exact LOCO	1.27	1.24	1.13	2.27	0.98	9.8	0.67	0.20
BOLT-LMM-inf	1.28	1.24	1.16	1.98	0.99	8.3	1.00	0.20
exact LOCO + PC1	1.11	1.09	1.10	1.13	1.14	1.0	0.67	0.20

in the neighbouring block. On the same data, the LD split lowers the r^2 leaking across boundaries by 13–14% and removes all S1 false loci (Table 3). At $n = 4,000$ it removes 6 of LOCO’s 17 false loci in the first seed, but 9 of the remaining 11 still sit next to a causal block. In this constant-size coalescent, $r^2 > 0.8$ reaches more than 1,000 SNPs between low-frequency variants, so no cut avoids every strong pair; for the same reason, the bigsnpr default of forbidding cuts through pairs with $r^2 > 0.3$ was infeasible.

3.5 Large samples and computation

At $n = 10,000$ the K-free variance-component fit (stochastic Lanczos quadrature on the streaming kinship) reproduced the exact fit’s δ to 0.09% (0.91724 against 0.91639), with no $n \times n$ matrix formed. A scan on a truncated top- k spectrum with a flat tail, which mixmogam had used for large n , was anti-conservative ($\lambda_{GC} = 1.30$ at $n = 10,000$ with $k = 1,024$); it is no longer a default, and large samples use the two-step statistics. Table 4 lists measured wall times.

Table 3: Scenarios S1 and S3 on the same genotypes and traits, cut into blocks every 200 SNPs (fixed) or by the optimal LD split: λ_{GC} under the split, false loci per scan and locus-level power under both. Means over three seeds.

scenario	method	λ_{GC}	false loci		power	
			fixed	split	fixed	split
S1 ($n = 800$)	no LOCO	0.95	0.67	0.00	0.10	0.12
	exact LOCO	1.25	1.33	0.00	0.20	0.22
	BOLT-LMM-inf	1.24	1.00	0.00	0.18	0.20
S3 ($n = 4,000$)	no LOCO	0.91	4.00	3.67	0.33	0.33
	exact LOCO	1.65	15.67	12.33	0.51	0.51
	BOLT-LMM-inf	1.65	16.00	12.33	0.47	0.50

Table 4: Measured wall times on a development laptop (Apple silicon, four BLAS threads, AC power), which compare mixmogam with itself; the per-SNP reference is a port of the 2010 code. S3 timings are means over three seeds.

workload	result
batched EMMAX scan vs. per-SNP least squares ($n = 2,000$, $m = 10^4$)	0.27 s vs. 33.7 s (123 $\times$)
exact scan throughput, $n = 2,000$, $m = 10^5$	48×10^3 SNPs/s
exact scan throughput, $n = 5,000$, $m = 2 \times 10^5$	7.8×10^3 SNPs/s
S3 ($n = 4,000$, $m = 5 \times 10^4$): exact LOCO, 25 groups	221 s
S3: BOLT-LMM-inf	18 s
S3: exact non-LOCO scan after one fit	2.7 s

4 Discussion

Single-constant calibration is a statement about the average SNP, and under population structure the average hides a gradient. On real *A. thaliana* genotypes the reference LDAK-KVIK program and mixmogam’s BOLT-LMM-inf are anti-conservative for SNPs that load weakly on structure, about 1.7 times the nominal FPR at $\alpha = 1\%$, and conservative for SNPs that load strongly on it, while their genome-wide λ_{GC} looks normal. The gradient follows from (5), so it applies to every statistic built that way, including those of REGENIE (Mbatchou et al., 2021). We measured it only at $\alpha = 1\%$ per quintile, and only in the programs we could run; its size at genome-wide thresholds and in the reference BOLT-LMM and REGENIE programs is not established here. The structure-aware denominator is principled for BOLT-LMM-inf. For the mixture and elastic-net statistics its transfer is a heuristic, and through LDAK-KVIK’s scaling rule it left about 7% overall inflation under strong simulated structure.

The S2 results restate an old caution in measured terms: a kinship-only mixed model cannot carry a strong environment on one ancestry axis, and a principal-component covariate can (Price et al., 2010; Yang et al., 2014; Sul et al., 2018). Loh et al. (2015) simulated an environment on the top principal component that explained 1% of the variance and found the mixed-model statistics calibrated; S2’s 50% is far beyond that. The S2 results add a second caution. Comparing LOCO with non-LOCO by λ_{GC} alone misleads under structure, because the non-LOCO scan’s deflation can cancel residual confounding.

Two simulation designs that look natural produced misleading results. A confounder placed on the leading eigenvector of the tested SNPs’ own kinship is a function of those SNPs: the SNPs that load on it most carry the confounder themselves, and only a kinship that contains them absorbs it.

LD blocks cut at fixed intervals of a contiguous segment leak QTL tags across “chromosomes”, which inflates false loci and leaks into LOCO groups. An independent ancestry axis and an LD-optimal split address both. Unlinked blocks, as in S2, remove the leakage entirely.

Limitations. The real data come from one inbred plant panel with strong structure and $n = 1,307$. The only reference program we ran is LDAK-KVIK; BOLT-LMM and REGENIE binaries were not available on this platform. Our two-step statistics are re-implementations, with documented departures. The simulations are small (three seeds per scenario, six replicates for calibration), the infinitesimal background makes some false loci real, and the environment in S2 explains 50% of the variance, which is a stress test rather than a typical cohort.

What would change these conclusions. Three findings would. First, a flat loading gradient for the reference BOLT-LMM and REGENIE programs on the same RegMap phenotypes would confine the problem to the implementations studied here. Second, a flat gradient in a human multi-ancestry cohort would limit it to strongly structured samples. Third, a calibrated exact LOCO scan without principal components when the environment explains a few percent of the variance would mark S2 as a stress test only.

Data and code availability

mixmogam, the benchmark scripts and every result used here are in the mixmogam repository, with archives under `benchmarks/results/`:

- 20261003T081812Z-structure-calibration: simulated structure;
- 20261003T083746Z-kvik-reference: RegMap, including the LDAK program;
- 20261003T122520Z-sim-study: S1–S3 and the post-hoc checks;
- 20261003T001500Z-sim-study-large: $n = 10,000$.

Each archive holds a source snapshot, seeds and versions. `report/make_figures.py` rebuilds the figures and tables of this report from those archives.

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